UFV FORGE

A $100,000,000 self-sustaining computing hub for students, professors and business

Project Charter and Management Pre-Plan

CIS-385 Project Management, University of the Fraser Valley

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Submitted to: Kevin Renso

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1. Executive summary

UFV Forge is a real-world computing hub that connects students, professors and local businesses. It combines advanced labs (a cybersecurity range, a GPU/HPC cluster and an IoT lab) with UFV-owned servers where every student gets a personal sandbox and students run live websites and services, in the spirit of platforms such as pwn.college. A Connection Fund pays student fellowships, faculty chairs and industry matching. A diversified revenue model and a $20 million endowment are intended to let the hub cover its own operating costs, so it keeps growing after the federal funding ends.

The project is a one-time public investment, so it is judged on public value and operational self-sustainability, not on commercial payback. The $100,000,000 is allocated in section 4, the schedule runs to the September 1, 2030, deadline with a two-month buffer, and financial targets are staged so the plan stays realistic in the hub's opening years. A written UFV bridge commitment, not the project contingency, covers any operating shortfall during ramp-up.

2. The problem and the evidence

Employers want graduates with hands-on experience in security, cloud and AI, but students rarely get to operate real infrastructure or work on real client problems. Local businesses struggle to find talent and affordable technical help. Professors need modern facilities and industry partners for applied research.

Finding	Evidence	Source
Large, documented talent gap	A 2022 ICTC report, citing a 2021 (ISC)² study, counted about 123,696 cybersecurity professionals in Canada and a shortage of about 25,000. It also found weak communication between industry and academia, and a risk of talent being drawn to the United States (Herron and Quan).	Herron and Quan
Graduates are far below demand	The 2025 State of Cybersecurity in Canada report says Canada produces fewer than 4,000 graduates a year against demand above 25,000 (Lexpert). Estimates vary by source: Bell Cyber's president cites at least 50,000 unfilled roles (CP24), so our plan treats the gap as ranging from 10,000 to 50,000.	Lexpert; CP24
Threat level is high	Fortinet's 2026 threat and skills-gap reports, as covered by MobileSyrup, place Canada second in the world for cyberattacks, behind only the United States. They also found that 82 per cent of Canadian organizations had at least one breach in the past year, and that 47 per cent of IT leaders name the cybersecurity skills shortage as a top cause of breaches (Shankar).	Shankar

Finding	Evidence	Source
Proven model: pwn.college	ASU's platform is free and open, used in 145 countries, and students spent over 1.5 million hours on it in 2023 (Arizona State University, "Next Generation"). It offers over 1,000 hands-on challenges (Arizona State University, "Education"). It is supported by DARPA funding (Arizona State University, "American Cybersecurity Education"), which shows the model scales but depends on government money. UFV Forge adds a self-funding model.	Arizona State University
Industry funds university cyber hubs	Bell invested $1M in UNB's McKenna Institute, aiming for 150 job-ready graduates a year (University of New Brunswick, "Bell Invests"; CP24). UNB's Canadian Institute for Cybersecurity opened with IBM and $4.5M in funding (BetaKit), and the federal government committed up to $10M over five years to a Cyber Attribution Data Centre there (University of New Brunswick, "Federal Funding"). These are much smaller than $100M, so the plan must justify its scale carefully.	University of New Brunswick; BetaKit

3. The proposed solution

Advanced labs

A hands-on cyber range for security practice, a GPU/HPC cluster for AI and data work, and an IoT/embedded lab. Every lab runs real equipment, not simulations only.

Student-run servers & platform

Our own on-campus servers host a self-paced practice platform (inspired by pwn.college at Arizona State University). Every student gets a personal sandbox server, and students operate, secure and maintain live websites and services.

Connection Engine

Businesses post real problems; professors supervise; students build solutions for pay or credit. A dedicated Connection Fund pays student fellowships, faculty chairs and matches industry money.

Self-sustaining model

The hub earns income from hosting, memberships, sponsored projects, training and grants, backed by a $20 million endowment, so it keeps running after the federal money is spent. Section 5 sets out the revenue streams and a staged, conservative path to full cost coverage.

4. Budget

The federal government has entrusted our group with $100 million to help the University of the Fraser Valley become a more innovative and competitive institution. The goal is to develop modern facilities, improve technology, create opportunities for students, and strengthen partnerships with business and industry. To make sure the funding is used responsibly, the budget is divided into the categories below.

Category	Amount	Share
Hub facility	$26,000,000	26%
Advanced labs and infrastructure	$20,000,000	20%
Connection Fund	$22,000,000	22%
Sustainability endowment	$20,000,000	20%
Platform build and launch team	$5,000,000	5%
Project management office	$2,000,000	2%
Contingency fund	$5,000,000	5%
Total	**$100,000,000**	**100%**

The largest portion, $26 million, builds the UFV Forge hub facility, including laboratories, studios and a server room where students gain practical experience with modern technology. A further $20 million provides the advanced laboratory equipment and infrastructure needed for research and development. The Connection Fund receives $22 million for fellowships, faculty positions and partnerships with outside organizations, creating opportunities for students to work alongside professionals. Another $20 million goes into a sustainability endowment that produces income to support the hub over the long term. The $5 million contingency fund is reserved for unexpected costs such as construction delays, equipment price increases or additional technology requirements. It is used only for project costs and is never used to cover operating deficits.

Operating costs after construction

Once construction is complete, the hub must be financially sustainable. UFV Forge has an estimated annual operating cost of about $5.5 million:

Operating cost	Per year
Staff salaries and employment costs	$3,000,000
Power, equipment and hardware	$1,500,000
Security	$500,000
Student operations	$500,000
Total	**$5,500,000**

Cost control

Managing the $100 million will be an important responsibility throughout the project. First, fixed-price contracts will be used whenever possible, so contractors and suppliers agree to a set price before starting work. This reduces the risk of unexpected cost increases. Second, monthly financial reviews compare planned spending with actual spending and the amount of work completed (earned value), so budget problems are found early. A cost performance index below 0.90 triggers a formal review. Third, spending from the contingency fund requires steering committee approval, and contingency is released only for project cost and schedule risks, never for operating deficits.

Financial performance is also reviewed at the three kill points described in section 12. If costs rise far above plan or progress falls behind, project leaders may adjust, delay or cancel certain activities.

5. Self-sustaining business model and return on investment

5.1 Revenue streams

To cover its operating costs, UFV Forge plans to generate about $6 million per year once fully operational. These are planning estimates and will be refined in the first year of the pilot.

Revenue stream	Annual estimate
Industry memberships (about 25 partners at roughly $60,000)	$1,500,000
Sponsored projects and student consulting	$1,200,000
Research grants and contracts	$1,000,000
Hosting services for nonprofits and small businesses	$800,000
Endowment income (about 4% of $20 million)	$800,000
Training programs and certifications	$700,000
Total	**$6,000,000**

If the targets are met, the hub produces a small annual surplus of about $0.5 million, which is reinvested in equipment, facilities and future student projects.

5.2 Staged targets and a conservative scenario

The hub opens in August 2030, so it cannot earn full revenue in 2030. The platform pilot (from December 2027) provides early income, but most revenue begins after opening. Our base-case target is a stretch goal: income covering 80% of operating cost on an annualized basis at opening and 100% by 2032. Because that is optimistic, we plan on a conservative scenario with a slower ramp-up.

Year	Base case coverage	Conservative coverage	Conservative shortfall
2030 (Aug.–Dec.)	80% (stretch)	40%	$1,375,000
2031	90%	50%	$2,750,000
2032	100%	70%	$1,650,000
2033	100%	85%	$825,000
2034	100%	100%	$0
Total			**$6,600,000**

The 2030 row covers the five months of operation after opening, about $2.3 million of operating cost. In the conservative scenario the ramp-up shortfall totals about $6.6 million over four years. If coverage is below 40% at the end of 2031, the steering committee reviews the service mix and costs. This risk is recorded in section 14.

UFV bridge commitment

The project contingency is reserved for project cost and schedule risks and is never used to cover operating deficits. Instead, the UFV Board provides a written bridge commitment of up to $7 million to cover any operating shortfall from 2030 to 2033, repaid from later operating surpluses where possible. The commitment is a condition of the project and must be signed before kill point 1 in June 2028. If it is not signed, kill point 1 cannot be passed.

5.3 Return on investment: public value, not commercial payback

A simple payback calculation must be stated honestly. At a surplus of about $0.5 million per year, recovering $100 million would take roughly 200 years, so UFV Forge is not a commercial investment. It is a one-time public investment whose return is measured in public value and operational self-sustainability: the hub covers its own running costs, preserves the $20 million endowment, and does not need ongoing federal operating funding beyond the temporary UFV bridge during ramp-up.

As an illustration, if the $80 million of capital spending (excluding the preserved endowment) is spread over a 30-year useful life, the cost is about $2.7 million per year. If the hub serves 1,000 students per year (a planning assumption), that is roughly $2,700 per student per year, before counting contributions from business partners.

Public value measure	Planning target	How it is measured
Students using hands-on labs and servers	1,000 per year by 2032	Platform and lab usage records
Active industry partners	25 by 2030	Signed membership agreements
Real client projects completed	40 per year by 2032	Project register
Paid student fellowships	100 per year	Connection Fund records

Public value measure	Planning target	How it is measured
Operating cost coverage	See staged targets	Annual financial review

6. Project charter

Project name	UFV Forge Computing Hub
Sponsor	Federal government (funding); UFV Board and President (delivery sponsor)
Project manager	Ranjit Gautam
Purpose	Create a real-world computing hub that links students, professors and business, trains job-ready graduates, and sustains itself financially.
Objectives (SMART)	Open and formally accept the hub before Sep 1, 2030 (target Aug 2030); launch the practice platform pilot by Dec 2027; sign 25 industry partners by Dec 2030; cover 100% of operating cost by 2032 (base case) or 2034 (conservative case); finish within the $100M budget.
Scope: included	Facility, labs, servers and network, platform development, hiring, partnership program, Connection Fund, revenue services, endowment.
Scope: excluded	Changes to degree programs, general campus IT, residences.
Deliverables	Hub building; cyber range; GPU/HPC lab; IoT lab; student server platform; partner portal and Connection Fund rules; business plan; PMO documents; final benefits report.
Success measures	1,000 students per year by 2032; 25 active partners; 40 real client projects per year; 95% platform availability; no reportable privacy breaches; cost coverage per the staged targets.
Key milestones	Charter approved Mar 2027; platform pilot Dec 2027; design review (KP1) Jun 2028; construction complete Dec 2029; systems review (KP2) May 2030; final acceptance (KP3) and opening Aug 2030.
Assumptions	Land or building available on campus; strong local industry interest; stable hardware prices; qualified staff can be hired; UFV's Board signs the bridge commitment before kill point 1.
Constraints	Fixed $100M budget; Sep 1, 2030, hard deadline; public procurement and privacy rules; project contingency may not be used for operating deficits.
Kill points	Three kill points (details in section 12): (1) after building and technical design, before construction; (2) after major technical systems are installed and tested; (3) at close-out, before final acceptance and opening.

6.1 Acceptance criteria and scope validation

Scope validation is the formal acceptance of completed deliverables by the sponsor. It is different from quality control, which checks that work is correct. Every deliverable has acceptance criteria agreed in advance. A deliverable is submitted, reviewed against its criteria, and then accepted or rejected in writing with reasons. Formal sign-off takes place at the kill points.

Deliverable	Acceptance criteria	Evidence	Accepted by
Hub building	Occupancy permit issued; meets BC Building Code; power and cooling commissioned for design load	Inspection reports, permit	UFV Facilities; Board delegate
Cyber range	Logically isolated from UFV's network; logging active; user agreements in place	Independent security test	UFV IT Security
GPU/HPC lab	Passes burn-in and benchmark tests; holds temperature at full load	Test reports	Technical lead; faculty sponsor
IoT lab	All equipment inventoried, tested and bookable	Inventory and test sheet	Faculty sponsor
Student server platform	Every student can get a sandbox; passes load test for 500 concurrent users; privacy impact assessment approved	Load-test report; PIA	UFV IT; Privacy Office
Partner portal and Connection Fund	Rules approved; first call for projects issued; 25 partners signed by Dec 2030	Board approval; agreements	Steering committee
Business plan and revenue services	Services priced; contract templates ready; first-year budget approved; UFV bridge commitment signed	Signed-off plan	Finance lead; UFV Finance
PMO documents and benefits report	Baselines, change log, lessons learned and final benefits report delivered	Documents	Project sponsor

7. Stakeholders

Stakeholder	Interest	Engagement	Influence
UFV students	Skills, jobs, access to real hardware	Advisory panel, surveys, student ops team	Medium
UFV professors	Research, teaching tools, funding	Faculty working group, chairs program	Medium
UFV staff / IT	Security, workload, support	Joint security and operations plan	Medium
UFV leadership & Board	Reputation, financial risk	Steering committee, monthly report	High
Local businesses and startups	Talent, solutions, innovation	Industry advisory board, membership tiers	High
Federal government / Prime Minister	Return on public funds, skills	Quarterly progress and audit reports	High
BC government / Premier	Tech workforce, regional growth	Joint announcements, grant alignment	High
City of Abbotsford, community	Local jobs, nonprofits served	Hosting for nonprofits, public events	Medium
Local First Nations	Land, consultation, participation	Early consultation and partnership	Medium
Employees and contractors	Contracts, safety, fair pay	Clear roles, weekly meetings	Medium
Instructor (Kevin)	Quality of plan and pitch	Final presentation (get the project)	High

8. Work breakdown structure (WBS)

Work package	Level 2 tasks
1 Project management	1.1 Charter and governance; 1.2 Schedule and budget control; 1.3 Risk and change management; 1.4 Quality and communications; 1.5 Closeout
2 Facility	2.1 Design; 2.2 Permits; 2.3 Construction; 2.4 Commissioning
3 Labs and infrastructure	3.1 Cyber range; 3.2 GPU/HPC cluster; 3.3 Server room and network; 3.4 Security hardening
4 Student platform	4.1 Pilot in existing space; 4.2 Sandbox servers; 4.3 Challenge content; 4.4 Hosting service
5 Connection Engine	5.1 Industry advisory board; 5.2 Partner portal; 5.3 Fellowships and chairs; 5.4 Matching fund rules

Work package	Level 2 tasks
6 Sustainability	6.1 Business plan; 6.2 Revenue services launch; 6.3 Endowment policy; 6.4 Annual financial review
7 Hiring and procurement	7.1 Hiring plan; 7.2 RFPs; 7.3 Contracts

8.1 Requirements management

Requirements are gathered from students, professors, UFV IT, business partners and regulators, and recorded in requirements register. A traceability matrix links each requirement to a deliverable, a test and an acceptance criterion, so nothing is built without a reason and nothing required is missed. Requirements are baselined at kill point 1, and any later change goes through the change control process in section 14.

8.2 Procurement

Major purchases follow UFV's public procurement rules, which the group will confirm with UFV Purchasing. Construction is procured through a competitive request for proposals using a fixed-price contract. Servers, GPUs and networking equipment are bought through competitive bids scored on price, technical fit, security and support. Time-and-materials contracts are used only for software development and only with a spending cap. All contracts include privacy and security clauses, warranties and clear acceptance tests, and the procurement plan avoids dependence on a single vendor wherever practical.

9. Technical requirements

For our proposal of UFV FORGE to be successful, the facility would need to incorporate the proper technical infrastructure to support cybersecurity, AI, data work, IoT development, and student-run services.

One of the main parts of the facility would be the cybersecurity range. The range will need to be operated in a controlled environment so that students are able to practise cybersecurity activities without affecting UFV's systems. The Government of Canada has guidance on Zero Trust which includes security controls such as multi-factor authentication, least privilege access, continuous monitoring, and encryption (Shared Services Canada). These controls would be useful for FORGE as students will be working with real systems and security tools.

The AI and high-performance computing lab will also require more computing power than a regular computer lab. The lab will need powerful computers, fast networking, storage, cooling, and enough electrical capacity to support the AI, data processing and research workloads, which are heavier.

The IoT lab will require development computers, sensors, microcontrollers, networking equipment, and proper testing equipment. Another essential part will be student-run servers, which would allow students to have their own sandbox environments and to operate any websites and services that they have created.

For these systems to stay protected, monitoring will be essential. The Government of Canada provides guidance explaining that event logging can help detect security incidents and provide information about what may have happened during an attack (Treasury Board of Canada Secretariat). This gives FORGE a basis for monitoring and logging so that technical or security issues in the student-run systems can be identified quickly. The cybersecurity range should stay separated from UFV's regular network so that students can safely practise attacks and defence without creating any risk for the university's normal systems.

10. Legal and building requirements

As FORGE is a new university facility, there are building requirements that need to be followed for the design and construction in British Columbia. The BC Building Code sets minimum requirements for areas such as safety, fire protection, structural protection, accessibility, and building efficiency (Province of British Columbia). These must be considered during the design phase, before construction begins.

Another set of requirements applies to the electrical component of the facility. FORGE will contain servers, networking systems, high-performance computers, IoT devices and other technology. As the building will be equipped with powerful equipment, it will need enough power and cooling capacity to support the equipment safely.

Another essential area is privacy. As FORGE is a university facility, it will have access to student logs, research data, and information from business projects. British Columbia's Freedom of Information and Protection of Privacy Act apply to public bodies and requires them to properly manage personal information (Office of the Information and Privacy Commissioner for BC). Because of this, access to sensitive information must be limited to authorized users, and the systems should be designed with privacy and security in mind.

11. Time management and schedule

The final deadline set for the project is September 1, 2030. To meet this deadline, the work will need to be completed in a proper sequence of steps with progress monitored throughout the project. In class, we have learned that project time management includes defining activities, sequencing, estimating resources and durations, developing the schedule, and controlling the schedule.

For UFV FORGE, the project would begin with planning and determining the requirements for the building and the technical design. Once those requirements are set and clear, the building and technical design can be completed. After this design is approved, the contracts and suppliers will be selected and construction will begin.

The technical equipment for the facility will be installed in a later phase of the project. This would include the student servers, networking systems, cybersecurity range, AI and high-performance computing lab, and IoT equipment. Once everything is installed, the systems will go through a full test phase so that the facility can officially open.

The phases must be completed in an order where certain things are in place before others can start. For example, the technical equipment cannot be installed until the building structure is complete, and final testing cannot begin until the equipment has been installed. These are

mandatory dependencies, because one activity must be completed before the next can move forward properly.

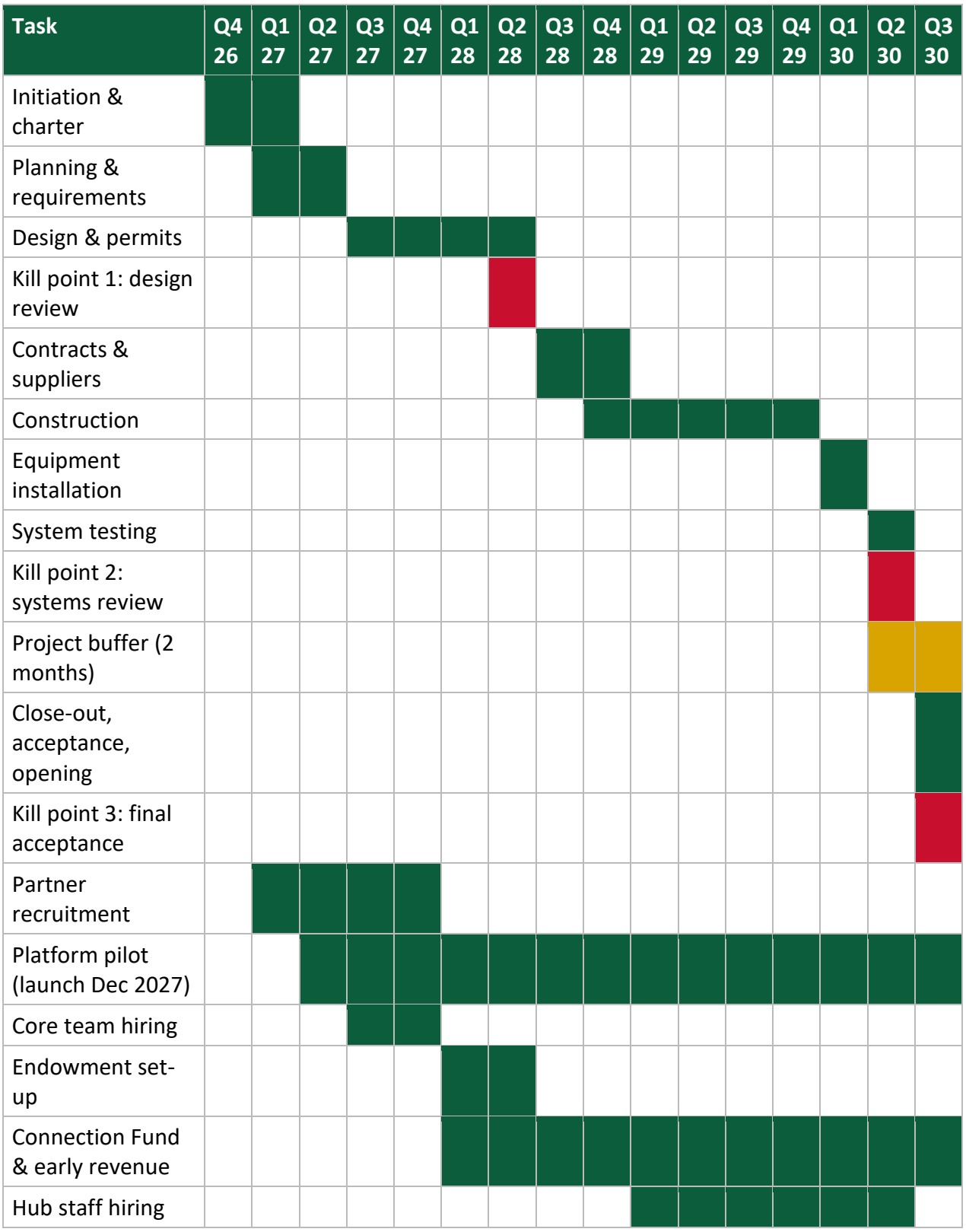
11.1 Phase schedule

The phases above are placed on a calendar below. Dates are planning dates from the project start in November 2026 to the September 1, 2030, deadline.

Phase	Start	End	Months	Depends on
Initiation and charter	Nov 2026	Mar 2027	5	None
Planning and requirements	Jan 2027	Jun 2027	6	Charter approved
Building and technical design, permits	Jul 2027	Jun 2028	12	Requirements
Kill point 1: design review	Jun 2028	Jun 2028	-	Design complete
Contracts and supplier selection	Jul 2028	Oct 2028	4	Kill point 1
Construction	Nov 2028	Dec 2029	14	Contracts signed
Equipment installation	Jan 2030	Mar 2030	3	Building structure complete
System testing	Apr 2030	May 2030	2	Equipment installed
Kill point 2: systems review	May 2030	May 2030	-	Testing complete
Project buffer	Jun 2030	Jul 2030	2	Absorbs delays on the critical path
Close-out, final acceptance, opening (kill point 3)	Aug 2030	Aug 2030	1	Kill point 2
Partner recruitment	Feb 2027	Oct 2027	9	Charter approved
Platform pilot (launch Dec 2027)	May 2027	Jul 2030	39	Charter approved
Core team hiring	Jul 2027	Dec 2027	6	Partner commitments
Endowment set-up	Jan 2028	Jun 2028	6	Pilot results
Connection Fund and early revenue services	Jan 2028	Aug 2030	32	Pilot results
Hub staff hiring	Jan 2029	Jun 2030	18	Construction under way

11.2 Gantt chart

Red = kill point; gold = schedule buffer. Each column is one quarter.



11.3 Critical path and schedule buffer

The critical path is planning and requirements, design and permits, contracts, construction, equipment installation, system testing, then close-out. These are the mandatory dependencies described above. Using PERT three-point estimates (expected duration = (optimistic + 4 x most likely + pessimistic) / 6), the expected critical-path duration is about 1.8 months longer than the planned duration, so the two-month buffer covers the expected delay but not the pessimistic case. That is why kill point 1 and kill point 2 are real go/no-go decisions.

Critical activity	Optimistic	Most likely	Pessimistic	Expected	Planned
Design and permits	10	12	18	12.7	12
Contracts and suppliers	3	4	6	4.2	4
Construction	12	14	20	14.7	14
Installation and testing	4	5	8	5.3	5
Total (months)	29	35	52	36.8	35

12. Kill points

For a project of this size, it is essential to have points where UFV can stop and review the project before moving into the next major stage. In class, we have learned that phase exits and kill points are used to decide whether a project should continue, be redirected, or be stopped.

Kill point 1 happens after the building and technical design has been completed, right before construction begins. The project team reviews the design, expected cost, timeline, and technical requirements. If any major problems appear, the direction of the project can be changed.

Kill point 2 happens once the major technical systems have been installed and system testing has been completed. The cybersecurity range, student servers, networking systems, AI and high-performance computing lab, and IoT lab are reviewed. Any major technical or security problems must be addressed before the project continues towards completion.

Kill point 3 (final) happens during the project close-out phase. UFV does its final review of the complete facility and decides whether the agreed work has been completed before officially accepting FORGE and opening it to students.

Kill point	Timing	Go criteria	Possible decisions
1. Design review	Jun 2028	Design approved; permits issued or on track; cost estimate within 5% of baseline; at least 10 partners committed; UFV bridge commitment signed	Continue, redirect scope, or stop

Kill point	Timing	Go criteria	Possible decisions
2. Systems review	May 2030	All systems pass tests; independent security test passed; no open critical defects	Continue, extend testing using buffer, or stop
3. Final acceptance	Aug 2030	All acceptance criteria met; handover documents delivered; first-year operating budget approved	Accept and open, or hold acceptance

13. Team roles and responsibilities

Role	Name	Responsibility
Project lead, web designer and presenter	Ranjit Gautam	Charter, integration, stakeholders, website, final report, presentation and video presentation
Finance lead	Pavneet Gill	Budget, revenue model, cost control, financial risks
Legal and time-projection lead	Gary Rai	Technical, legal and time management: requirements, schedule, Gantt chart, kill points
Team member	Abiola Azeez	Video clips and final editing support

14. Controlling costs, risks and changes

Change control

Any change to scope, cost or schedule starts with a written change request describing its impact on scope, cost and time. The project manager reviews it, and approves changes under $250,000 that do not affect a kill-point date. Larger changes go to the steering committee. Approved changes are logged and the baselines are updated.

Risk register

Risk	Rating	Owner	Trigger	Response
Security breach on hub servers	High	Gary Rai	Failed security test or any reportable incident	Isolated networks, security staff, regular audits, incident plan
Revenue below plan	High	Pavneet Gill	Coverage under 40% at end of 2031, or revenue 30% below plan for two quarters	Review service mix and costs; draw on the UFV bridge commitment (never the project contingency)

Risk	Rating	Owner	Trigger	Response
Weak industry uptake	Medium	Ranjit Gautam	Fewer than 10 partners committed by kill point 1.	Pilot first; advisory board; early sponsorship commitments.
Permit or design delay	Medium	Gary Rai	Permits not issued by Jun 2028.	Start consultation early; use schedule buffer; escalate to steering committee.
Construction cost inflation	Medium	Pavneet Gill	Cost performance index below 0.90.	Fixed-price bids; contingency; value engineering.
Hiring shortage	Medium	Ranjit Gautam	Key roles unfilled 6 months before opening.	Recruit a year early; student-staff pipeline.
Privacy incident or non-compliance	Medium	Gary Rai	Privacy impact assessment not approved before launch.	Limit access; privacy-by-design; Privacy Office review.
Scope creep	Medium	Ranjit Gautam	More than 3 unapproved change requests in a month.	Formal change control; sponsor sign-off.

15. Quality, communication and lessons learned

15.1 Quality management

Quality is planned, not inspected in at the end. The quality standards are the BC Building Code, BC privacy law and Government of Canada security guidance such as Zero Trust. Quality assurance checks that processes are followed through design reviews, audits and security reviews. Quality control checks the work itself through building inspections, equipment burn-in tests, load tests and independent security testing. Quality measures include inspection defects closed on time, platform availability of at least 95%, security findings closed within 30 days, and cost and schedule performance indexes.

15.2 Communication plan

Audience	Message	Frequency	Method	Owner
Steering committee and sponsor	Status, cost, schedule, risks, decisions	Monthly	Report and meeting	Ranjit Gautam
Federal government	Progress and audit information	Quarterly	Written report	Ranjit Gautam

Audience	Message	Frequency	Method	Owner
Project team	Tasks and issues	Weekly	Team meeting	Ranjit Gautam
Contractors	Site progress and safety	Weekly	Site meeting	Gary Rai
Finance reviews	Budget vs. actual, earned value	Monthly	Finance report	Pavneet Gill
Students and staff	Updates, feedback	Monthly	Website and advisory panel	Ranjit Gautam
Industry partners	Opportunities and results	Quarterly	Advisory board meeting	Ranjit Gautam

15.3 Lessons learned

Lessons are recorded in a lessons-learned register at each kill point and at close-out, noting what happened, why, and what to do differently. The register is shared with UFV and partners. The plan already applies lessons from comparable projects: pwn.college depends on government funding, so UFV Forge adds a self-funding model, and the UNB hubs are far smaller than $100 million, so revenue targets are staged and a conservative scenario is included.

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